

Program 3 : Code a dockerized python flask or Node.js Application

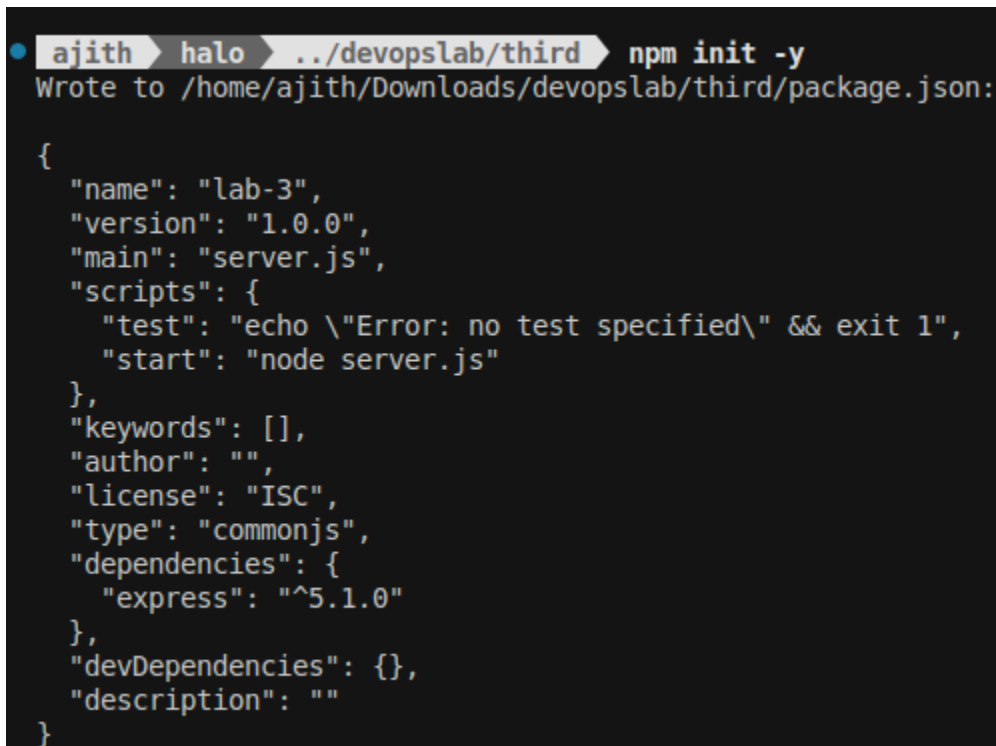
Project Structure

- |__Dockerfile
- |__Server.js
- |__package.json
- |__node-modules

Steps to proceed with the program

Step 1: Initialize the Node.js project folder

\$ npm init -y



```
ajith@halo:~/devopslab/third$ npm init -y
Wrote to /home/ajith/Downloads/devopslab/third/package.json:

{
  "name": "lab-3",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "description": ""
}
```

Step 2: Install express to run as a server

```
$ npm install express
```

```
ajith@halo:~/devopslab/third$ npm install express
up to date in 54s

16 packages are looking for funding
  run `npm fund` for details
```

Step 3: Run the docker build command

```
$ sudo docker build -t third .
```

```
ajith@halo:~/devopslab/third$ sudo docker build -t third .
[sudo] password for ajith:
[+] Building 1.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 160B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 103.39kB
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY package*.json ./
=> CACHED [4/5] RUN npm install
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> writing image sha256:a1cf51e6f7ec9deb1918ed4d3753693f3740cb05284ffcf616a674199df2d6ea
=> => naming to docker.io/library/third
```

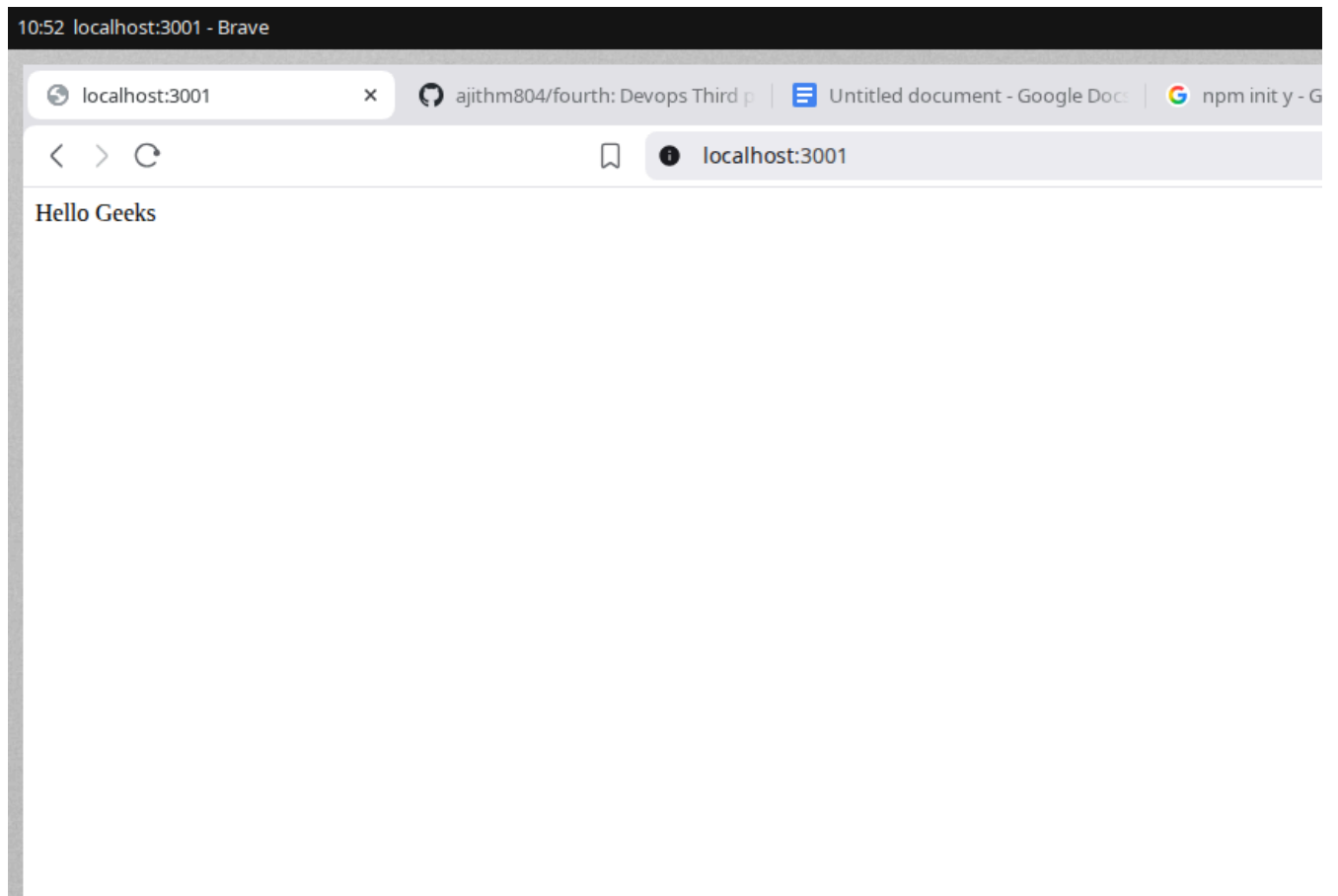
Step 4: After that try executing the docker image with the command

```
$ sudo docker run -d 3001:3001 third
```

```
ajith@halo:~/devopslab/third$ sudo docker run -p 3001:3001 third
> lab-3@1.0.0 start
> node server.js

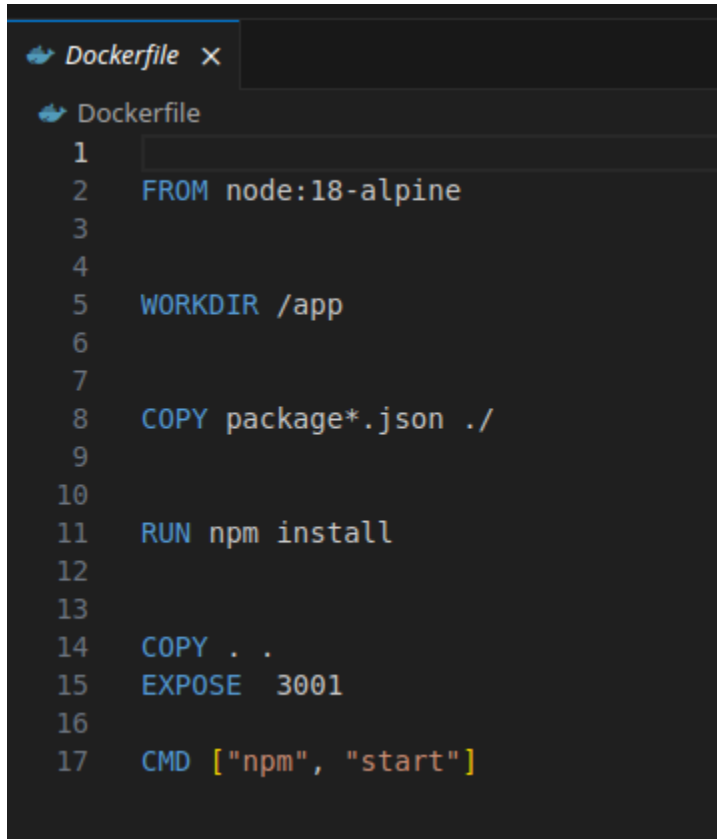
Server listening on 3001
```

Step 5: With the above command getting executed, open browser and navigate to <http://localhost:3001>



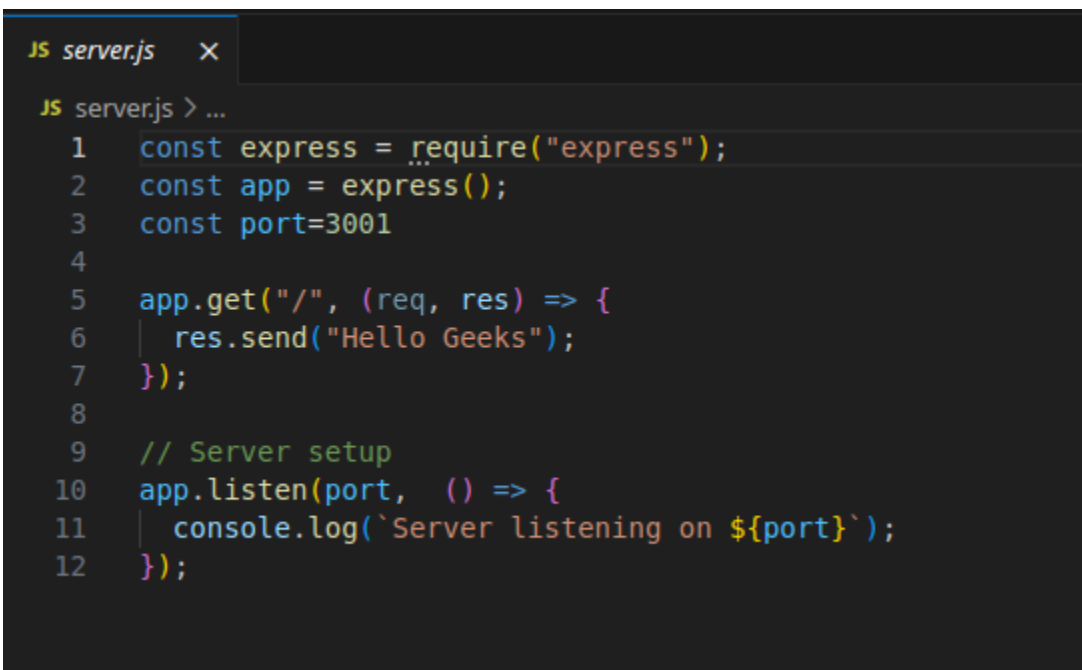
Project Files:

1. Dockerfile

A screenshot of a code editor showing a Dockerfile. The file has a tab labeled 'Dockerfile' with a Docker logo and a close button. The content of the Dockerfile is as follows:

```
1  
2 FROM node:18-alpine  
3  
4  
5 WORKDIR /app  
6  
7  
8 COPY package*.json ./  
9  
10  
11 RUN npm install  
12  
13  
14 COPY . .  
15 EXPOSE 3001  
16  
17 CMD ["npm", "start"]
```

2. server.js

A screenshot of a code editor showing a file named 'server.js'. The file has a tab labeled 'JS server.js' with a close button. The content of the file is as follows:

```
JS server.js > ...  
1 const express = require("express");  
2 const app = express();  
3 const port=3001  
4  
5 app.get("/", (req, res) => {  
6   res.send("Hello Geeks");  
7 });  
8  
9 // Server setup  
10 app.listen(port, () => {  
11   console.log(`Server listening on ${port}`);  
12 });
```

3. package.json

```
{ } package.json x
{ } package.json > ...
1  {
2    "name": "lab-3",
3    "version": "1.0.0",
4    "main": "server.js",
5    "scripts": {
6      "test": "echo \"Error: no test specified\" && exit 1",
7      "start": "node server.js"
8    },
9    "keywords": [],
10   "author": "",
11   "license": "ISC",
12   "type": "commonjs",
13   "dependencies": {
14     "express": "^5.1.0"
15   },
16   "description": ""
17 }
18
```

4. Project Structure as in IDE

